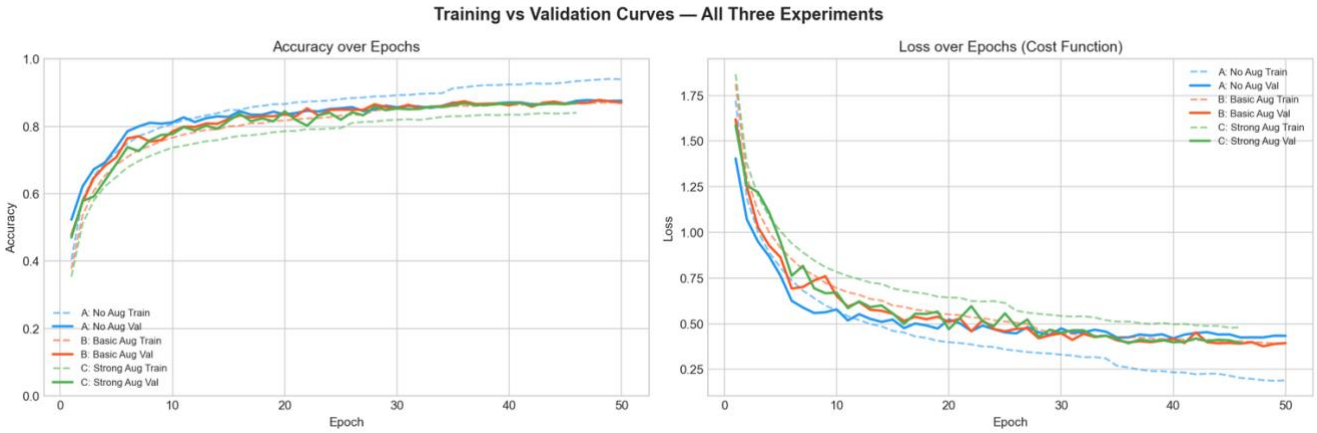
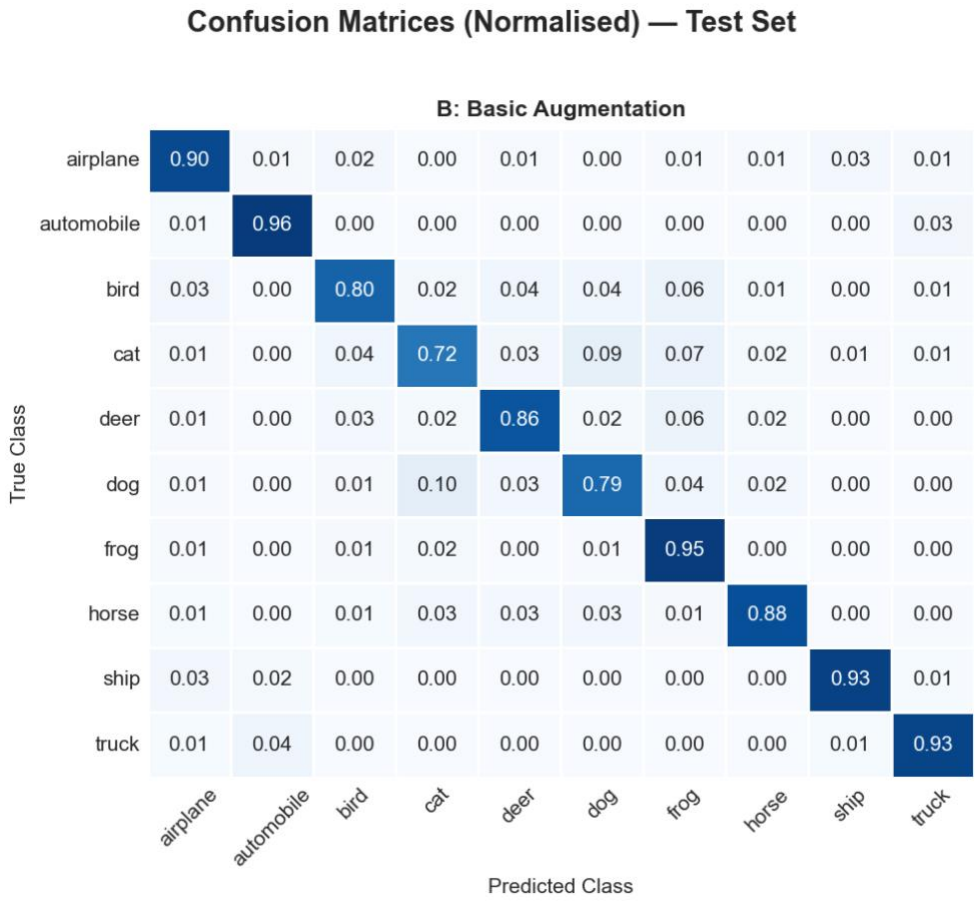
Artefact 3: Independent Project – CNN for Object Recognition (CIFAR-10)

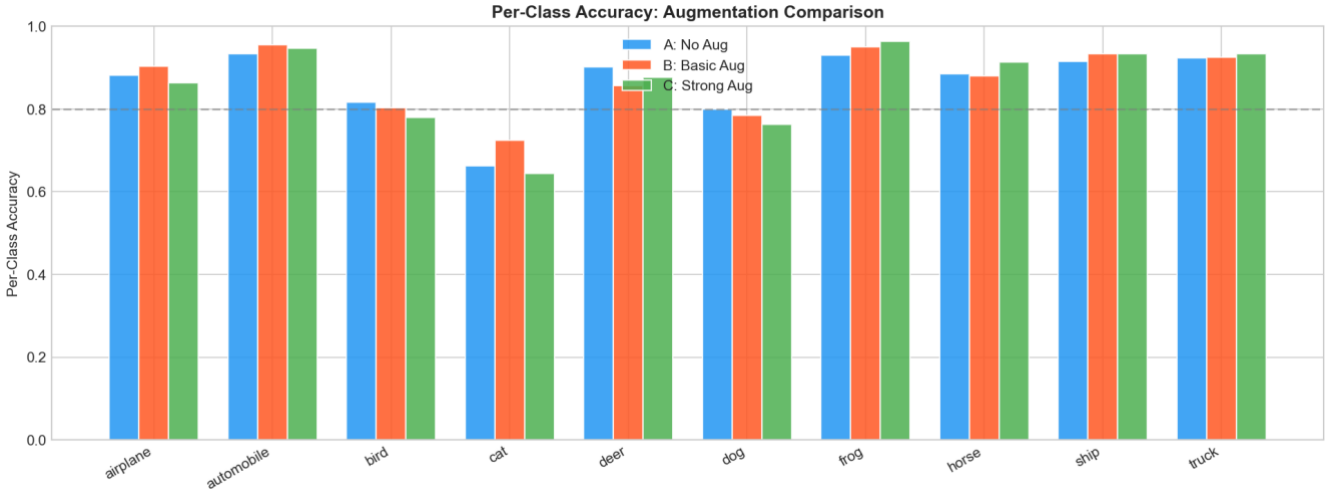
CNN Presentation with Slide Show : [cifar10CNN_presentation_FINAL.ppsx]
Training curves : accuracy vs validation for experiments A, B, C)



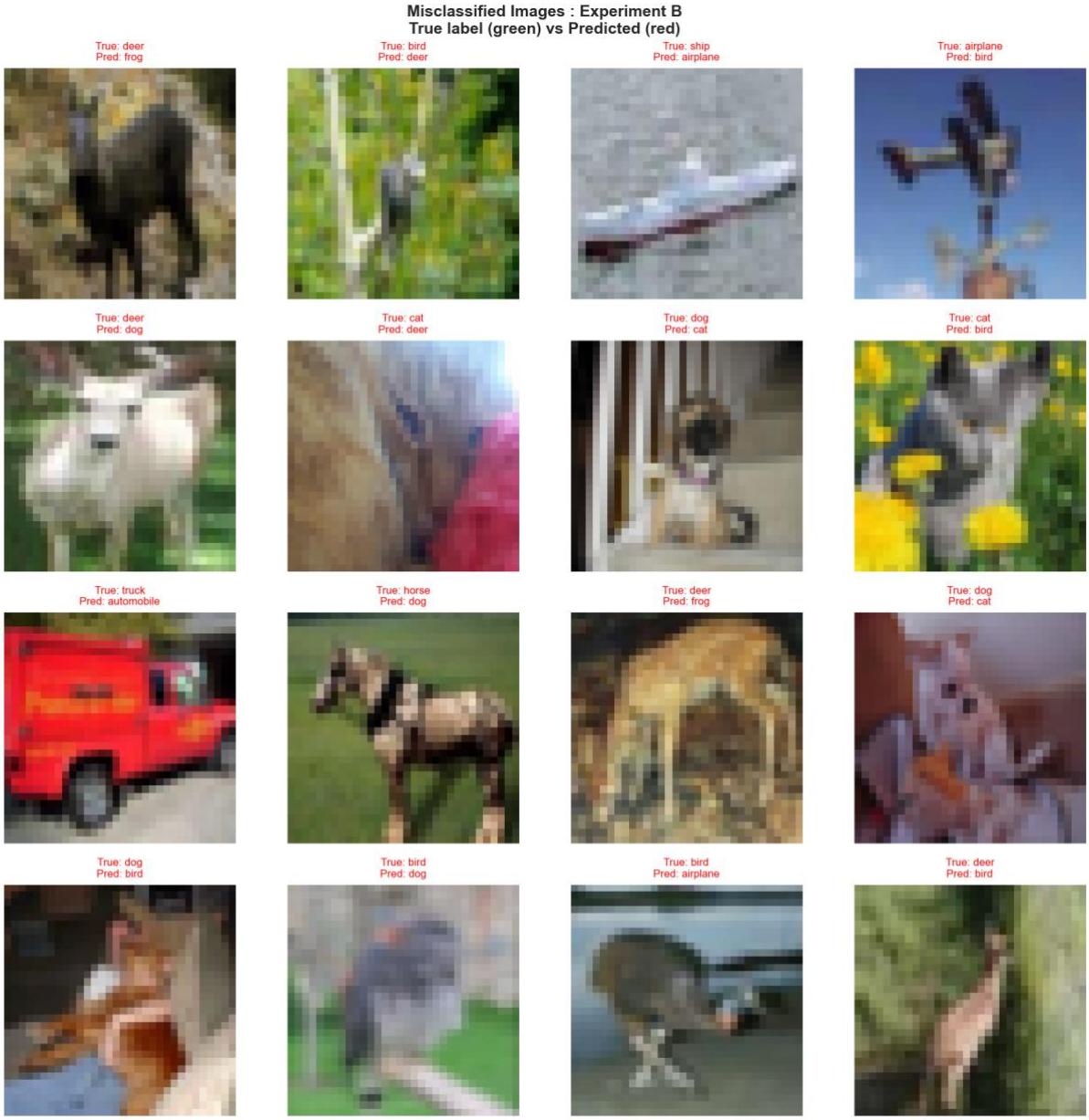
Confusion matrix for best model – Experiment B



Per-class accuracy comparison chart



Example misclassifications for Experiment B: error analysis images



This artefact represents my independent implementation of a convolutional neural network (CNN) for image classification using the CIFAR-10 dataset. The project focused on designing, training, and evaluating a deep learning model while systematically analysing the impact of data augmentation on model performance.

This work demonstrates my ability to apply and critically evaluate more advanced machine learning techniques beyond the core module content. I designed a controlled experiment comparing three configurations: no augmentation, basic augmentation, and strong augmentation. By keeping all other variables constant, I was able to isolate the effect of augmentation on model generalisation.

The results showed that basic augmentation achieved the best performance (87.17% test accuracy) with minimal overfitting, while no augmentation led to clear overfitting and strong augmentation introduced excessive noise. This demonstrated that model performance depends not only on architecture, but also on the balance between data variability and signal preservation.

A key strength of this artefact is the critical evaluation of model behaviour. Through confusion matrices and error analysis, I identified consistent misclassification patterns, particularly between visually similar classes such as cats and dogs. This highlighted that some errors were due to inherent dataset limitations, such as low image resolution and class similarity, rather than model failure.

The project also reinforced important machine learning principles, including the role of preprocessing, the importance of separating training, validation, and test datasets to avoid leakage, and the need to evaluate models beyond accuracy alone. Metrics such as overfitting gap and per-class performance provided a more complete understanding of model quality.

Overall, this artefact demonstrates my ability to design experiments, implement deep learning models, and critically interpret results, while also recognising the practical limitations of machine learning systems in real-world scenarios